

4.5 Equilibria

Knowledge of the concepts introduced in Unit 2, *Topic 2.9: Chemical equilibria* will be assumed and extended in this topic.

Students will be assessed on their ability to:

- a demonstrate an understanding of the term *dynamic equilibrium* as applied to states of matter, solutions and chemical reactions
- b recall that many important industrial reactions are reversible
- c use practical data to establish the idea that a relationship exists between the equilibrium concentrations of reactants and products which produces the equilibrium constant for a particular reaction, e.g. data on the hydrogen-iodine equilibrium
- d calculate a value for the equilibrium constant for a reaction based on data from experiment, e.g. the reaction of ethanol and ethanoic acid (this can be used as an example of the use of ICT to present and analyse data), the equilibrium $\text{Fe}^{2+}(\text{aq}) + \text{Ag}^{+}(\text{aq}) \rightleftharpoons \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$ or the distribution of ammonia or iodine between two immiscible solvents
- e construct expressions for K_c and K_p for homogeneous and heterogeneous systems, in terms of equilibrium concentrations or equilibrium partial pressures, perform simple calculations on K_c and K_p and work out the units of the equilibrium constants
- f demonstrate an understanding that when ΔS_{total} increases the magnitude of the equilibrium constant increases since $\Delta S_{\text{total}} = R \ln K$
- g apply knowledge of the value of equilibrium constants to predict the extent to which a reaction takes place
- h relate the effect of a change in temperature on the value of ΔS_{total} .